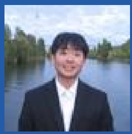

Python Lists vs Sets vs Tuples

Data with Dylan | Cheat Sheet | Python Fundamentals #4

WHAT ARE LISTS, SETS & TUPLES?

Collections store multiple values inside one variable, but each collection follows different rules.

💡 Think of a list as an editable playlist, a set as a bag of unique items, and a tuple as a sealed coordinate pair.

CORE DATA & FEATURES

COLLECTION TYPE	SYNTAX EXAMPLE	MUTABLE? (YES/NO)	ORDERED? (YES/NO)	ALLOWS DUPLICATES?	BEST USED FOR
List	`["red", "blue"]`	**Yes**	**Yes**	**Yes**	Editable sequences that need indexing or a meaningful order.
Set	`{"red", "blue"}`	**Yes**	**No**	**No**	Unique values, removing duplicates, and fast membership checks.
Tuple	`(10, 20)`	**No**	**Yes**	**Yes**	Fixed records or coordinates that should not change.

KEY SYNTAX & OPERATIONS

Access ordered items

```
colors = ["red", "blue", "green"]
colors[0]       # -> "red"
point = (10, 20)
point[1]        # -> 20
```

Add or change mutable collections

```
colors.append("yellow")
colors[1] = "navy"
fruits.add("orange")
fruits.remove("banana")
```

Convert between collection types

```
unique = set([1, 1, 2])   # -> {1, 2}
fixed = tuple([1, 2])     # -> (1, 2)
again = list(fixed)       # -> [1, 2]
```

TYPE & FORMATTING WARNINGS

⚠ {} creates a dictionary; use set() for an empty set.

⚠ (5) is an integer; (5,) is a one-item tuple.

⚠ Sets cannot be indexed, and their displayed order may change.

⚠ True equals 1, so a set may treat them as duplicates.

MINI PRACTICE

Choose the best collection:
A. Unique user IDs with duplicates removed
B. Fixed (x, y) coordinates
C. An editable sequence of tasks

Hint: set, tuple, list - then try set([101, 101, 205]).

SET & TUPLE BEHAVIOR

Membership, length, and duplicate removal

```
fruits = {"apple", "banana"}
"apple" in fruits       # -> True
len({"apple", "apple"})  # -> 1
len({1, True, 2})       # -> 2
```

Build a new tuple instead of changing one

```
original = (1, 2, 3)
extra = (4, 5)
updated = original + extra
# -> (1, 2, 3, 4, 5)
```

COMMON BEGINNER MISTAKES

✗ **Indexing a set**

`fruits[0]  # TypeError`

✓ **`"apple" in fruits  # membership check`**

✗ **Creating an empty set**

`items = {}  # this is a dict`

✓ **`items = set()`**

✗ **Forgetting the tuple comma**

`single = ("apple")  # string`

✓ **`single = ("apple",)`**

✗ **Changing a tuple item**

`point[0] = 5  # TypeError`

✓ **`point = (5, point[1])`**

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